MILITARY SPECIFICATION

CYLINDERS, CARBON DIOXIDE, FOR INFLATION OF PNEUMATIC LIFE PRESERVERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers three types of puncturable-seal, carbon dioxide-filled cylinders.

* 1.2 Classification. Carbon dioxide cylinders shall be of the following types, as specified:

Type I - 25 to 28-gram capacity.

Type II - 28 to 31-gram capacity.

Type III - 34 to 37-gram capacity.

* 2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-C-101	-	Carbon Dioxide (CO_2): Technical and U.S.P.
QQ-P-416	-	Plating, Cadmium (Electrodeposited).
TT-I-1795	-	Ink, Marking, Stencil, Opaque, (Porous and Non-porous Surfaces).
PPP-B-566	-	Box, Folding, Paperboard.
PPP-B-576	-	Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585	-	Box, Wood, Wirebound.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio Air Logistics Center, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PPP-B-591	-	Boxes, Shipping, Fiberboard, Wood-Cleated.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Box, Wood, Nailed and Lock-Corner
PPP-B-636	-	Box, Shipping, Fiberboard.
PPP-B-640	-	Box, Fiberboard, Corrugated, Triple-Wall
PPP-B-676	-	Boxes, Setup

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-S-7742	-	Screw Threads, Standard, Optimum Selected Series: General Specification for.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U.S. Government Property.
MIL-STD-143	-	Standards and Specifications, Order or Precedence for the Selection of.
MIL-STD-794	-	Part and Equipment, Procedures for Packaging and Packing of.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1188	-	Commercial Packaging of Supplies and Equipment.

* (Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

CONSOLIDATED FREIGHT CLASSIFICATION COMMITTEE

Consolidated Freight Classification Rules and Regulations

(Application for copies should be addressed to the Official Classification Committee, Room 202, Union Station, 516 West Jackson Boulevard, Chicago IL 60606.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. This specification makes provisions for first article inspection.

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3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials.

3.3.1 Steel. The material shall be steel of a grade and composition suitable for the purpose intended.

* 3.3.2 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4 Design and construction.

* 3.4.1 Dimensions and shape. The type I cylinder shall conform in dimensions and shape to Figure 1. The type II cylinder shall conform to Figure 2. The type III cylinder shall conform to Figure 3.

* 3.4.2 Gas charge. Filled cylinders shall contain carbon dioxide conforming to grade B, type I of BB-C-101, as follows:

Type I - Not less than 25 nor more than 28 grams

Type II - Not less than 28 nor more than 31 grams.

Type III - Not less than 34 nor more than 37 grams.

3.4.3 Puncturable area. Each cylinder shall be provided with a puncturable area as shown on Figure 1 and 2. The puncturable area shall be either an integral part of the cylinder or shall be incorporated in a puncturable low-carbon steel sealing cap or disk securely welded to the cylinder.

3.5 Performance.

3.5.1 Leakage. Filled cylinders shall show no evidence of leakage at any point when subjected to the leakage and cyclic leakage tests specified in 4.5.3 and 4.5.7.

3.5.2 Deformation. Cylinders shall be examined to insure that there is no deformation of the cylinder or of the cylinder puncturable area after being tested as specified in 4.5.5.

3.5.3 Hydrostatic pressure. Cylinders shall not burst when subjected to an internal hydrostatic pressure of 8,000 pounds per square inch (psi), when tested as specified in 4.5.8.

3.5.4 Rupture. When the filled cylinder is subjected to the rupture test specified in 4.5.9, there shall be no splintering or fragmentation upon rupture of the cylinder. Complete separation of the puncturable area or of any portion of the cylinder assembly from the cylinder shall be considered as fragmentation.

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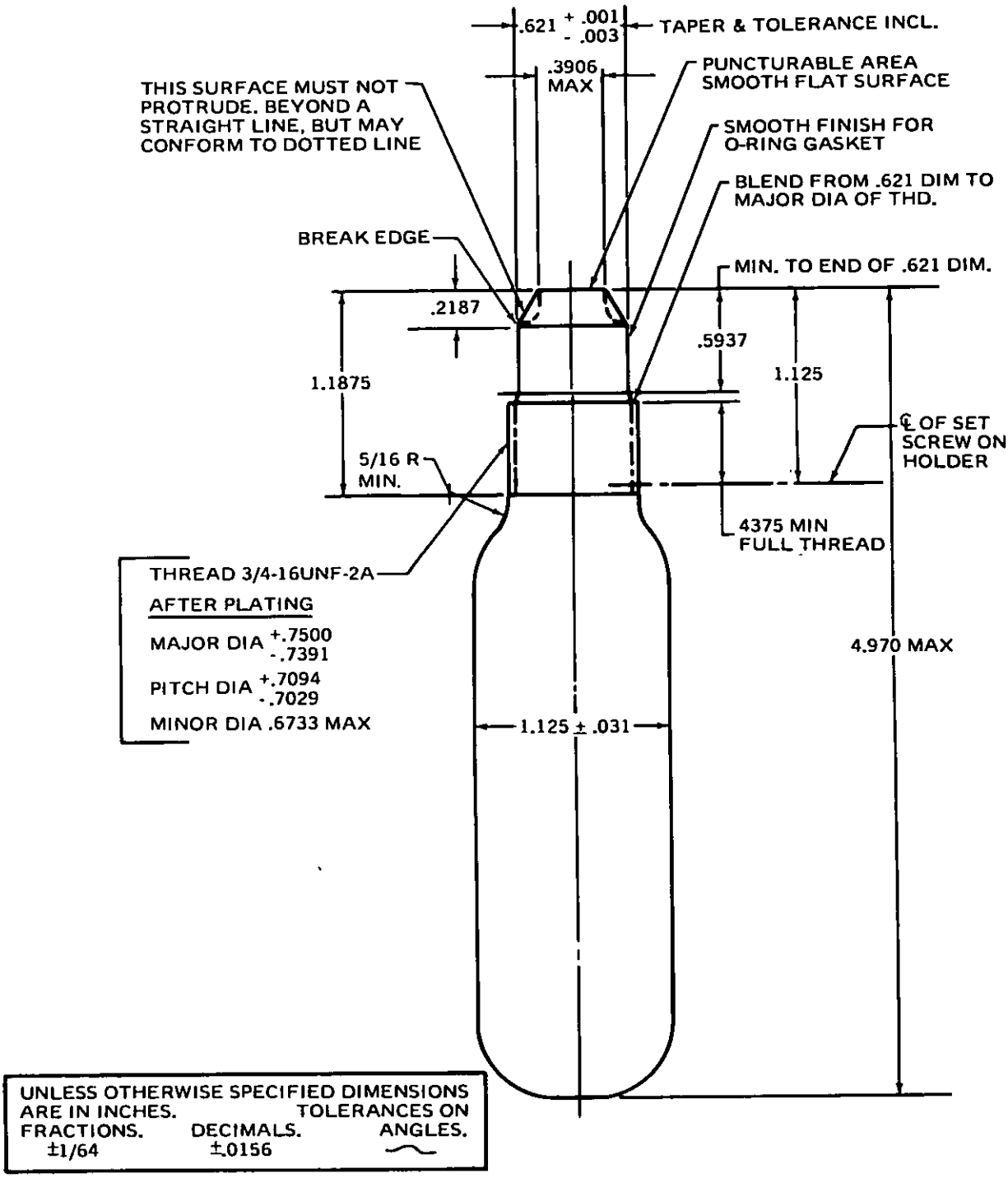


FIGURE 1. Carbon dioxide cylinder-type I.

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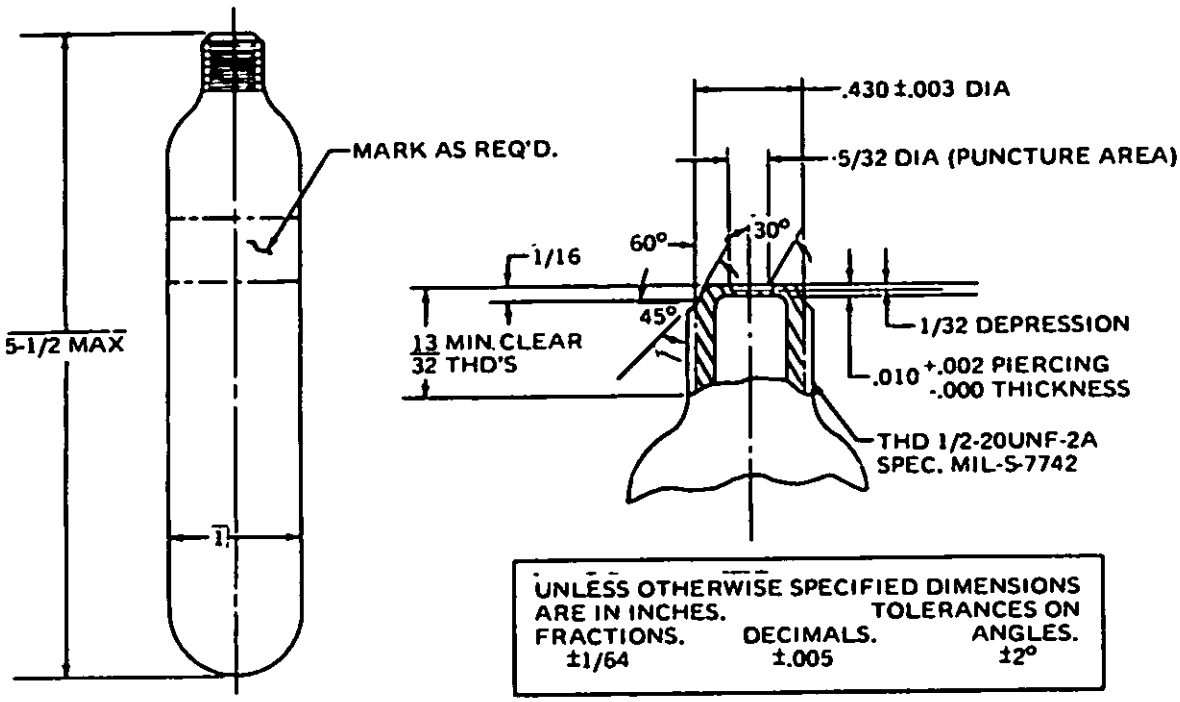


FIGURE 2. Cylinder-type II.

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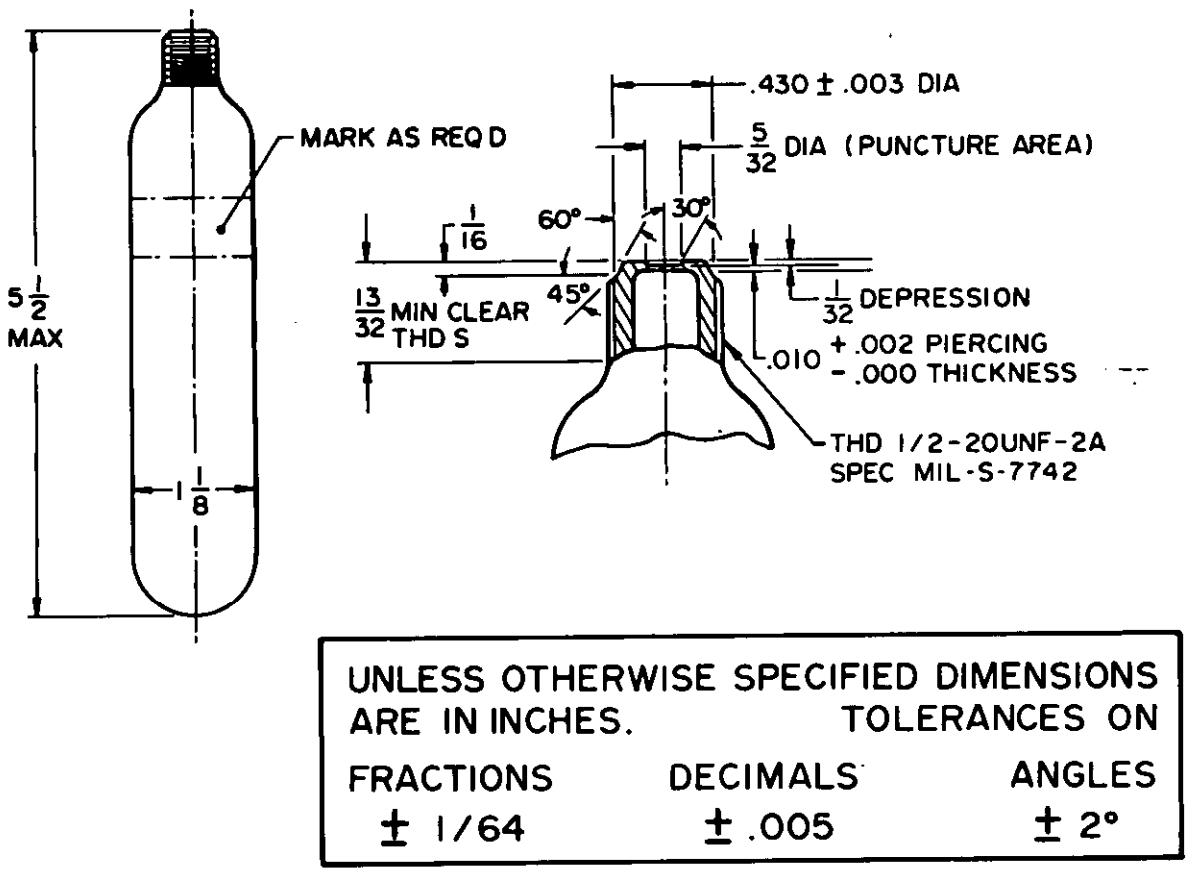


FIGURE 3. Cylinder-type III.

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* 3.5.5 Maximum filling density. The maximum filling density of the cylinder shall be 72 percent for type I and 76 percent for type II and type III. The filling density is the percent ratio of the weight of carbon dioxide in a container to the weight of water that the container will hold.

3.6 Finish. Cylinders, including threads, shall be cadmium plated in accordance with the requirements for type II, class 2 of QQ-P-416. The exposed surface shall be reasonably smooth, without any sign of bulging or deformation, and shall be free from pit marks and cracks.

* 3.7 Identification of product. The physical item shall be marked for identification in accordance with MIL-STD-130. The cylinder shall be marked with ink conforming to TT-I-1795, black or dark blue in color, and the marking area shall be coated with a compatible, clear lacquer. The markings shall consist of legible and durable numerals and letters not less than 1/8 inch high to indicate the following specific information:

CYLINDER, CARBON DIOXIDE, FOR INFLATION OF PNEUMATIC LIFE PRESERVES
MIL-C-25369
Applicable Type No.
Minimum Gross Weight Grams

Information to be entered by the manufacturer

3.7.1 Indicated minimum gross weight shall be the empty weight of the cylinder (rounded off to the next lower 1/2 gram), plus the weight of the minimum gas charge.

3.8 Workmanship. The cylinder shall be constructed in a thoroughly workmanlike manner. The finished cylinder shall be clean and free from oils, greases, and all other materials or defects that might affect appearance or functionability. The interior of the cylinder shall be free from rust, scale, or foreign matter.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of tests. The inspection and testing of the cylinders shall be classified as follows:

a. Quality conformance inspection (see 4.2).

b. First article inspection (see 4.3).

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4.2 Quality conformance inspection. Quality conformance inspection shall consist of:

a. Individual tests.

b. Sampling plans and tests.

4.2.1 Individual test. Each cylinder shall be subjected to the following tests as described under test methods:

a. Gas charge (see 4.5.1).

b. Leakage (see 4.5.3).

4.2.1.1 Dimensional inspection (gage) type I cylinders only. Inspection for dimensional (gage) inspection shall be in accordance with MIL-STD-105. Inspection level shall be S-4 with an acceptable quality level (AQL) of 0.15 percent defective (see 4.5.2).

4.2.2 Sampling plans and tests. The tests specified are considered as destructive tests, and the test samples shall not be submitted on the contract or order. Unless otherwise specified sampling for inspection shall be in accordance with MIL-STD-105. A lot for inspection purpose shall consist of all cylinders of one type presented for acceptance at one time.

4.2.2.1 Sampling plan "A." Cylinders shall be selected at random from each lot and subjected to the tests listed below and described in 4.5. The inspection level shall be S-4 with AQL 2.5 percent defective. (No less than 6 cylinders shall be selected.)

a. Examination of product (see 4.5.4).

b. Deformation (see 4.5.5).

c. Dimensional check (see 4.5.6).

d. Cyclic leakage (see 4.5.7).

4.2.2.2 Sampling plan "B." Three of the sample cylinders selected in accordance with sampling plan "A" shall be subjected to the hydrostatic pressure test, as described in 4.5.8.

4.2.2.3 Sampling plan "C." Three additional sample cylinders from sampling plan "A" shall be subjected to the rupture test, as described in 4.5.9.

4.2.2.4 Sampling plan "D." Six cylinders shall be selected at random from each lot of 2000 cylinders or fraction thereof and subjected to the maximum filling density test as described in 4.5.10.

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4.2.2.5 Rejections. Failure of any inspection sample to meet specified tests shall be cause for rejection of the entire lot from which the sample was selected. A lot which has been rejected on the basis of the hydrostatic pressure or rupture tests may be resubmitted for acceptance provided five other cylinders selected at random from the lot represented pass these tests. If any of the group of five fail to meet the test requirements, the lot represented shall be rejected and shall not be resubmitted for acceptance.

4.3 First article inspection.

4.3.1 First article test samples tested by the contractor. The contractor shall subject 12 cylinders of each type to the first article inspection specified in 4.3.2.

4.3.2 First article inspection. First article inspection shall consist of all tests described under 4.5.

4.4 Test conditions.

4.4.1 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 77° Fahrenheit (F) ±18°. Where tests are made with atmospheric pressure or temperature substantially different from the above values, proper allowance shall be made for the change in instrument reading.

4.5 Test methods.

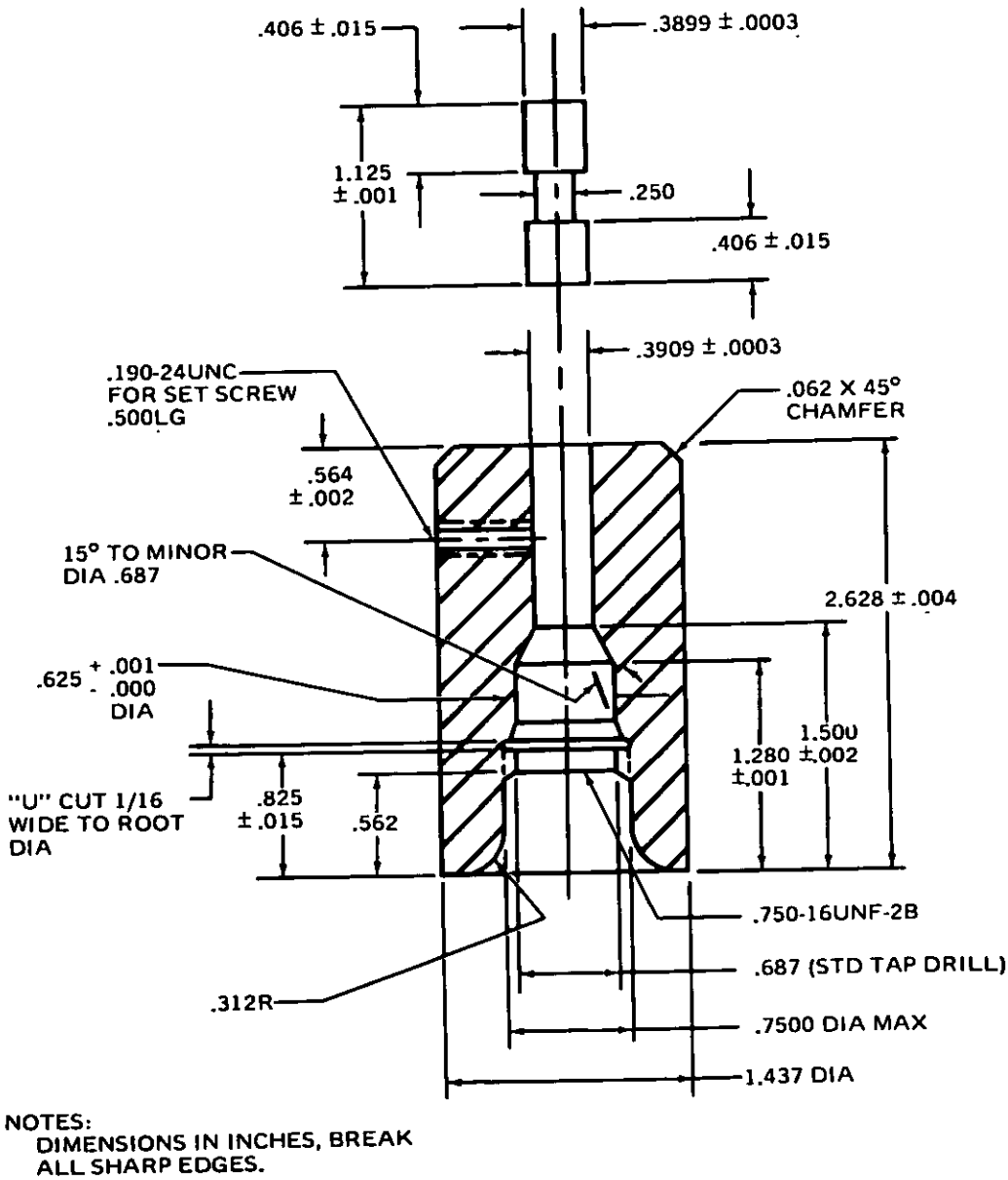
4.5.1 Gas charge. Each cylinder shall be weighed prior to charging with carbon dioxide. The cylinders shall then be charged, sealed, and heated to a temperature of +160°F ±2°, and maintained at this temperature for 20 minutes. After a period of 24 hours has elapsed, cylinders shall be weighed to determine compliance with 3.4.2, and failure to comply shall be cause for rejection.

4.5.2 Dimensional inspection (gage for type I cylinders only). Type I filled cylinders shall be dimensionally checked with a gage conforming to the cavity dimensions shown on Figure 4. Failure of the puncturable area of cylinder or seat firmly against bottom of gage cavity shall be cause for rejection.

4.5.2.1 Gage pin. The gage shall be fitted with a pin located on the axis of the gage at the center of the 0.3906 inch diameter (see Figure 4). The pin shall be used as a secondary indicator to assure that the cylinder has seated firmly against the gage cavity. The length of the pin shall be such that the end away from the cylinder will be flush with the outside surface of the gage by plus or minus 0.0003 inch when the cylinder is seated firmly against the bottom of the gage cavity.

4.5.3 Leakage. After the gas charge has been determined, filled cylinders shall be immersed in a water solution (1 percent aerosol OT clear, 25 percent, or equal) to a depth so that the uppermost portion of the cylinder is at least 1 inch below the water level for a minimum of 2 minutes. A 20 inch vacuum (mercury) shall be maintained above the surface of the solution.

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NOTES:
DIMENSIONS IN INCHES, BREAK
ALL SHARP EDGES.

FIGURE 4. Carbon dioxide cylinder gage for type I cylinder.

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4.5.4 Examination of product. Cylinder shall be visually inspected to determine conformance with the requirements of this specification with respect to general construction, finish, and workmanship. Failure of cylinders to meet these requirements shall be cause for rejection.

4.5.5 Deformation. There shall be no deformation of the cylinder or of the cylinder puncturable area after heating to a temperature of $+160^{\circ}F \pm 2^{\circ}F$, for not less than 20 minutes. Any evidence of deformation shall be cause for rejection.

4.5.6 Dimensional check. Sample cylinders shall be measured to determine compliance with the dimensional requirements of Figures 1 and 2. Dimensional checks shall be made following the leakage tests. Any dimensional variation in excess of the specified tolerances shall be cause for rejection.

4.5.7 Cyclic leakage. Filled cylinders shall be subjected to two complete temperature cycles. Each cycle shall consist of maintaining the cylinders at $-65^{\circ}F \pm 2^{\circ}F$ for 2 hours, and then at $+160^{\circ}F \pm 2^{\circ}F$ for 2 hours. After completion of the temperature cycles, cylinders shall be tested for leakage in conformance with 4.5.3 when specified (see 6.2).

4.5.8 Hydrostatic pressure. The sample cylinders shall be subjected to an internal hydrostatic pressure of not less than 8,000 psi at $+70^{\circ}F \pm 2^{\circ}F$ for a period of not less than 5 seconds. Bursting of cylinders under these conditions shall be cause for rejection. This test shall be conducted after the cylinders have been marked.

4.5.9 Rupture. The sample cylinders shall be ruptured by gradually heating in air. There shall be no splintering, fragmentation, or failure at the cap weld when the cylinders are ruptured. Complete separation of the puncturable area or of any portion of the cylinder assembly from the cylinder shall be considered as fragmentation, and shall be cause for rejection.

WARNING: This test is dangerous (see 6.3).

4.5.10 Maximum filling density. Sample charged cylinders shall be weighed, then the carbon dioxide discharged and cylinders weighed again. The difference between the first weighing and the second is the quantity of carbon dioxide inside. The cylinders shall then be filled with water and weighed. The ratio of the weight of water to the weight of carbon dioxide is the density, calculated as follows:

$$\text{Filling density} = \frac{\text{Weight of } CO_2}{\text{Weight of } H_2O} \times 100$$

4.6 Inspection of preparation for delivery. The preservation, packaging and packing shall be inspected to determine conformance to requirements of 5.1.

5. PACKAGING

* 5.1 Preservation and packaging. Preservation and packaging shall be A, C or industrial as specified (see 6.2).

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5.1.1 Level A.

5.1.1.1 Cleaning and drying. The cylinders shall be cleaned and dried in accordance with the requirements of MIL-P-116.

* 5.1.1.2 Unit packaging. Six cylinders of one type only shall be packaged method III in accordance with MIL-P-116 in folding or set-up box conforming to PPP-B-566 or PPP-B-676. The cylinders shall be separated by paperboard partitions and snuggy cushioned to prevent movement within the container.

* 5.1.2 Level C. The cylinder shall be cleaned and dried in accordance with the requirement of MIL-P-116. Preservation shall conform to the requirements of MIL-STD-794 for this level.

* 5.1.3 Industrial. Preservation shall be in accordance with the requirements of MIL-STD-1188.

* 5.2 Packing. Packing shall be Level A, B, C or industrial, as specified (see 6.2).

* 5.2.1 Level A. Packaged cylinders shall be packed in overseas type, style optional, grade B wood boxes conforming to PPP-B-601, or PPP-B-621.

* 5.2.2 Level B. Packaged cylinders shall be packed as specified for Level A except the box may be domestic type, or packed in overseas type boxes conforming to PPP-B-636, PPP-B-640, PPP-B-576, PPP-B-585 (class 3) or PPP-B-591.

* 5.2.3 Level C. Packing shall conform to the requirement of MIL-STD-794 for this level.

* 5.2.4 Industrial. Packing shall be in accordance with the requirements of MIL-STD-1188.

* 5.3 Marking.

* 5.3.1 Levels A, B and C. In addition to any special or other identification marking required by the contract (see 6.2), each unit pack, intermediate and exterior container shall be marked in accordance with MIL-STD-129.

* 5.3.2 Industrial. Industrial marking shall be in accordance with the requirements of MIL-STD-1188.

* 6. NOTES

* 6.1 Intended use. Cylinders covered by this specification are intended for use in inflating pneumatic life preservers.

* 6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Requirements for preproduction approval.

c. Type required (see 1.2).

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d. Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

e. The cyclic leakage test shall be conducted only when specified by the Government procuring activity (see 4.5.7).

6.3 Suggested equipment for rupture test. A perforated 3/8 inch thick steel container with bolted top and a can of sterno fuel is suggested.

6.4 First article reports and test samples for Navy procurement. First article inspection reports and samples for procurement for the Naval Air Systems Command shall be forwarded to the Air Crew Equipment Laboratory, Naval Air Engineering Center, Philadelphia PA.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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